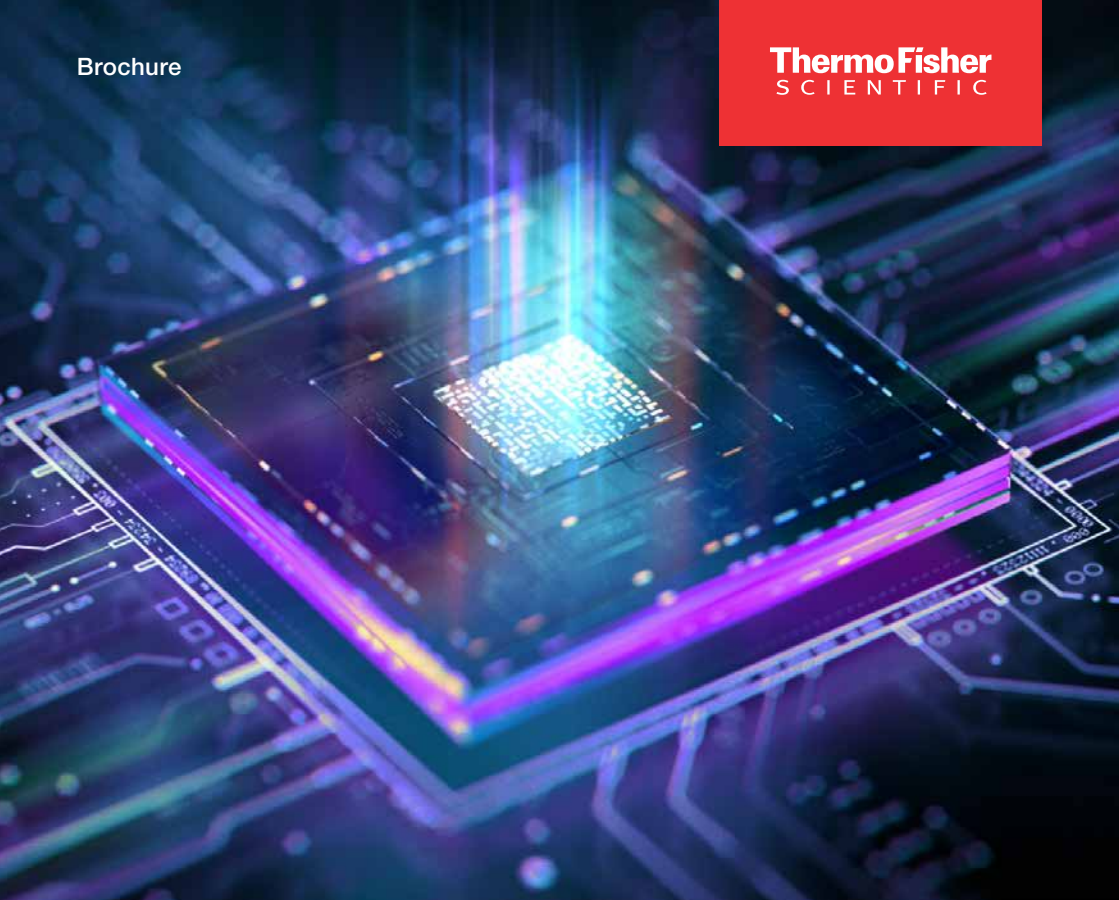




Semiconductor Solutions

About Thermo Fisher Scientific

Our Mission is to enable our customers to make the world healthier, cleaner and safer. Whether our semiconductor customers' products find use in the fields of finance, medicine, transportation, or education, we are here to support them with innovative solutions.

In order to reveal what matters, enable what's critical, and innovate what's next, we provide purpose-built electrical and physical analysis solutions that help enable lab and fab engineers to develop advanced semiconductor processes while improving yield, productivity, and cost-effectiveness. With over 40 years of semiconductor-focused innovation, we understand the unique challenges you face with increasingly complex technology nodes. We drive technological evolution for leading global companies while strengthening our expertise and capabilities to fulfill our semiconductor promise to each customer.

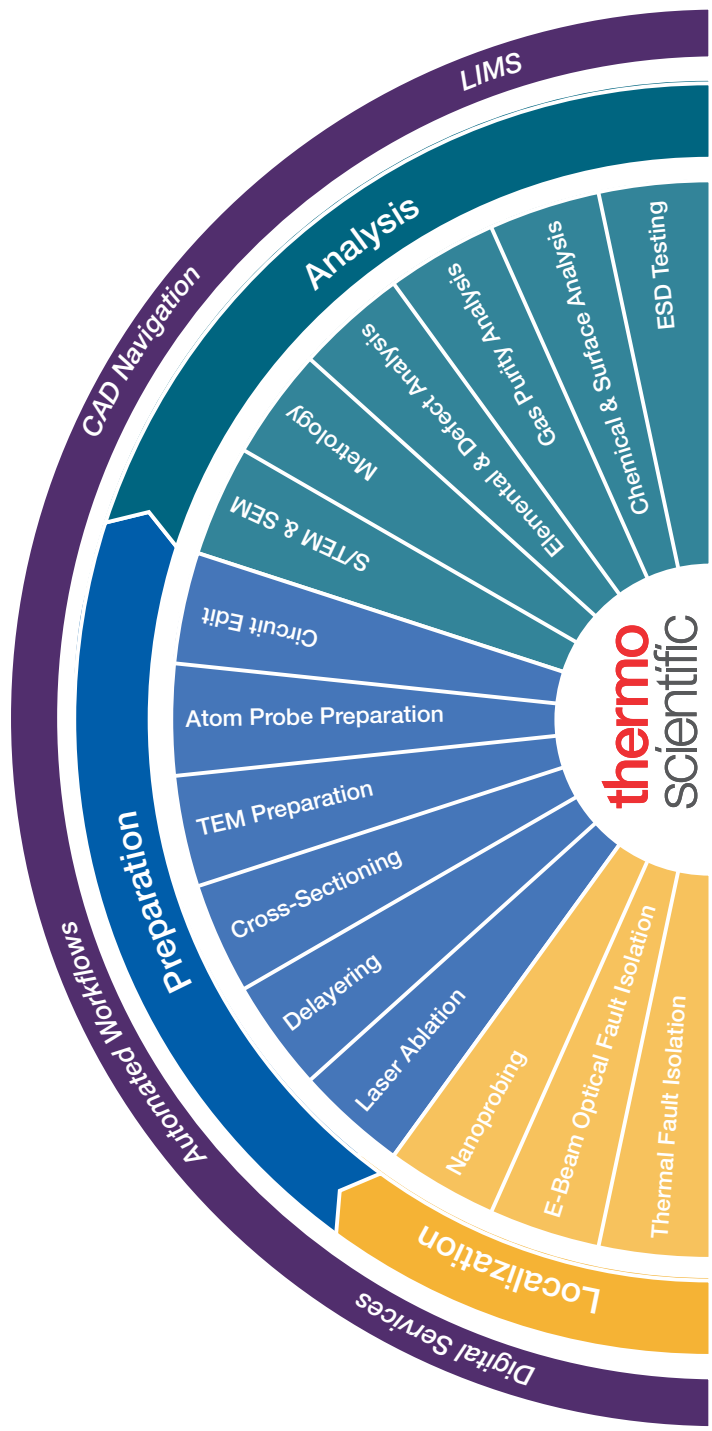
Our comprehensive portfolio of electrical fault isolation and characterization techniques, scanning electron microscopes (SEMs), transmission electron microscopes (TEMs), and focused-ion-beam (FIB)-SEMs work together as integrated solutions to reveal what is invisible to conventional techniques. These purpose-built technologies provide detection and analysis with atomic-scale accuracy for ground truth insights essential for complex 3D structures and advanced process development.

What sets us apart is our semiconductor-first approach. Each solution we develop addresses the specific metrology and defect challenges semiconductor customers face daily. Our advanced software suites and automated workflows enable fast time-to-data and operational productivity critical for staying competitive.

Our end-to-end comprehensive workflows combine these analytical instruments with dedicated semiconductor expertise to drive technological evolution and innovate what's next in semiconductor development. Our solutions provide electrical failure analysis, defect characterization, and metrology capabilities for next-generation nodes, delivering the faster, clearer path to higher yields and lower costs. With our extensive global support network and large global installed base, we're your trusted partner for semiconductor analysis, helping you see beyond the surface to gain the ultimate competitive advantage.

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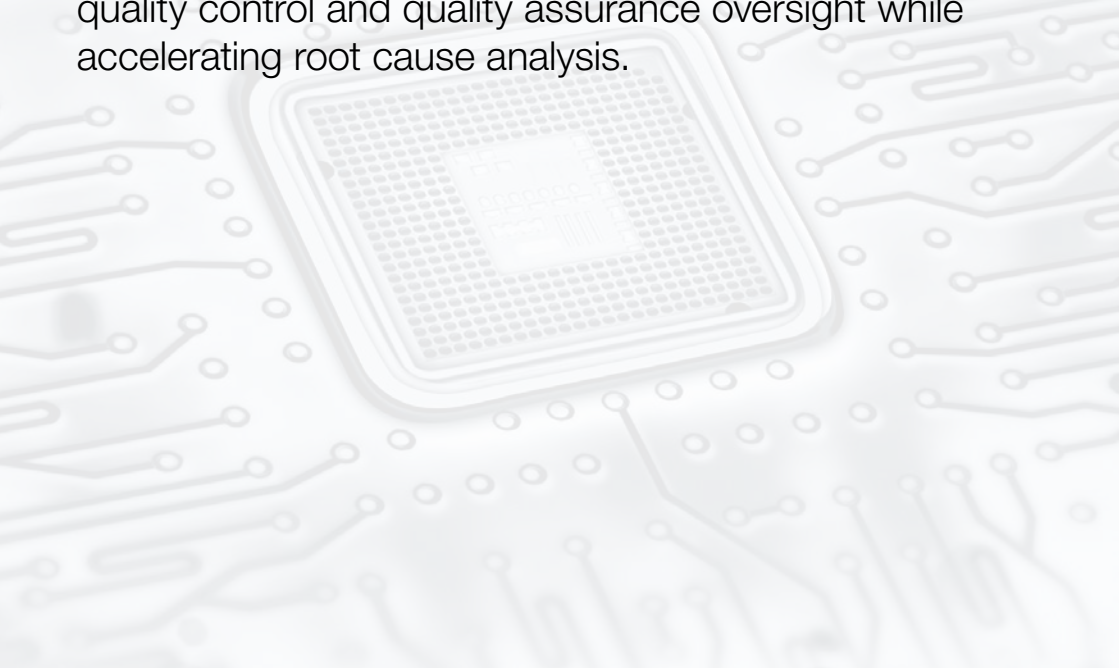
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Accelerating research and development • Solving complex analytical challenges • Increasing lab productivity and yield

Localization

For semiconductor engineers developing advanced processes and devices, the challenges of shrinking technology, new materials, and complex structures are creating defectivity issues. These non-visual defects manifest as electrical faults that compromise device performance, threaten long-term reliability, and erode yield targets. With today's semiconductor architectures featuring high-density interconnects, wafer-level stacking, flexible electronics, and integral substrates, failure-inducing defects have unprecedented hiding places, making accurate characterization increasingly difficult yet more critical than ever. Our purpose-built electrical failure analysis tools provide engineers with the localization and characterization information needed to identify subtle electrical issues, significantly enhancing quality control and quality assurance oversight while accelerating root cause analysis.



Localization

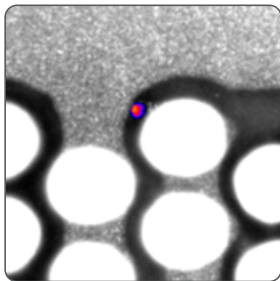
Electrical Analysis

Thermo Scientific™ ELITE™ System



Heat-based, non-destructive electrical fault localization at the die, package or wafer level.

- Fast detection and precise localization
- Thermal diagnostics and characterization inc. absolute temperature mapping
- 3D packaging solutions
- Optional OBIRCH, laser mark/scanning microscope
- 300 mm and ATE dockable configurations

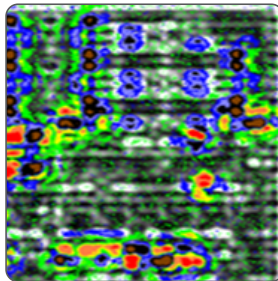


Thermo Scientific™ Meridian™ System



Electrical fault localization and circuit timing analysis at the die or wafer level, based on emission or laser stimulation.

- Dynamic analysis and laser voltage imaging/probing
- Sub-cell localization
- Higher signal noise ratio (SNR)
- Configurable to support power, memory or logic applications

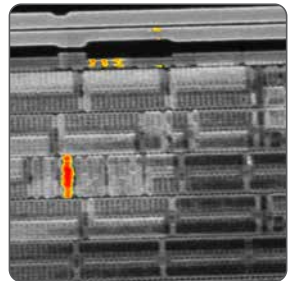


Thermo Scientific™ Meridian™ EX System



Innovative electron-beam-based solution for precise defect localization in advanced logic devices.

- Novel electron beam probing
- 10× resolution versus optical
- <5 nm node defect localization



Localization

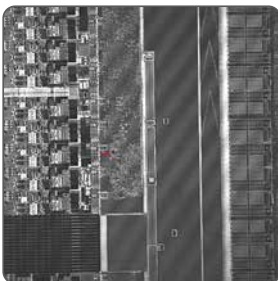
Electrical Analysis

Thermo Scientific™ Meridian™ S System



Static optical fault isolation for identifying systematic process, design or integration issues and root causing electrical failures.

- Fault Diagnostic with Active Probe Technology
- Static laser stimulation (SLS/OBIRCH) and photon emission options
- Supports micro-probing and probe card device stimulation

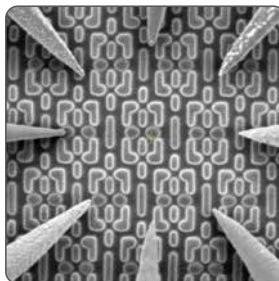


Thermo Scientific™ nProber™ IV System



SEM-based platform with comprehensive FEOL and BEOL fault localization and characterization.

- High-resolution, low-kV SEM for sensitive devices
- Low-noise electrical transistor measurements
- EBAC/EBIRCH BEOL metalization opens / shorts localization
- Fully integrated and automated probes

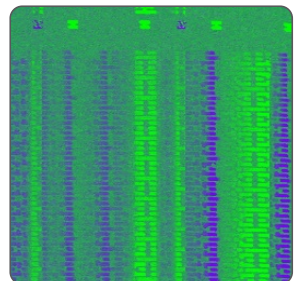


Thermo Scientific™ Hyperion™ II System



Atomic force-based nanoprobe for transistor fault localization and device characterization.

- Direct probe force and positional feedback
- Pico-current and SCM for fast fault localization
- Up to 8 probes for device characterization
- Highly-automated, ease-of-use



Preparation

Advanced physical and electrical analysis is required across semiconductor device types and technology nodes for yield management and process control. Precise, efficient sample preparation is a critical part of any workflow, with data fidelity and accuracy dependent on the quality of the sample preparation. We provide advanced sample preparation solutions based on our focused ion beam (FIB) and beam chemistry technologies, from leading automation solutions for high-volume sample production to exceptional multi-ion species platforms for optimal milling results. These solutions are most commonly used for creating cross-sections, S/TEM lamella, layered regions for nanoprobe, and atom probe tomography samples.



Preparation

Physical Analysis (*SEM, STEM, FIB Sample Prep*)

Thermo Scientific™ Helios™ 6 HD FIB-SEM



High-quality, ultra-thin with increased TEM sample prep productivity on the most advanced nodes.

- Next-gen FIB column designed specifically for TEM prep applications
- End-pointing accuracy and repeatability delivers high-quality TEM data

NEW!

Thermo Scientific™ Helios™ 6 HX FIB-SEM



Highly flexible FIB-SEM with *in situ* sample preparation, SEM and S/TEM imaging.

- Fully auto TEM sample prep
- Sub-nm targeting accuracy
- Robust FIB-SEM workflow

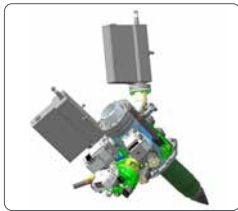
NEW!

Thermo Scientific™ Helios™ 6 HXS FIB-SEM

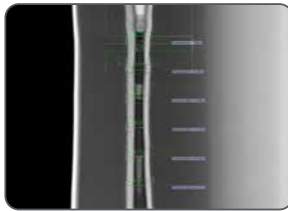


Highly flexible FIB-SEM with *in situ* sample preparation, SEM and S/TEM imaging.

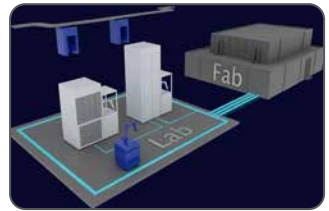
- Automated material handling
- Automated data connectivity
- Automated FIB-SEM workflows



Helios 6 HD FIB-SEM's Osprey FIB Column designed for high-quality sample preparation.



30 kV STEM image of a cross section of lamella showing tight thickness control.



Automated TEM sample prep within the Vulcan Automated Lab workflow.

Preparation

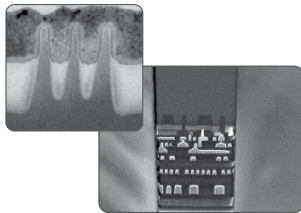
Physical Analysis (*SEM, STEM, FIB Sample Prep*)

Thermo Scientific™ Helios™ 5 FX DualBeam



Highly flexible FIB-SEM with *in situ* sample preparation, SEM and S/TEM imaging.

- Rapid time to sub-nm-scale data with 3 Angstrom resolution, in situ S/TEM
- Semi-automated TEM and atom probe sample preparation



Top: High-resolution *in situ* STEM image of a 14 nm device, cut across the fins of the device.

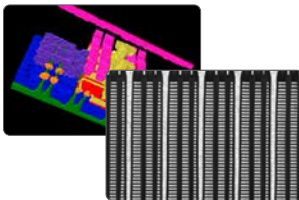
Bottom: Low-kV SEM image of a TEM lamella.

Thermo Scientific™ Helios™ 5 UC/CX DualBeams



Highly configurable for a wide range of high-resolution analysis applications.

- Flexible stage and loadlock configurations
- 150 mm/200 mm wafer capable
- Automated “slice and view” and reconstruction software
- Suitable for logic, memory, advanced power, analog and MEMS device analysis



Top: 3D reconstruction of 14 nm FinFET device produced with a Helios 5 DualBeam and Avizo Software.

Bottom: Low-voltage SEM cross-section of 3D NAND device produced with a Helios 5 UX DualBeam.

NEW! Thermo Scientific™ Scios™ 3 FIB-SEM



Fully featured for a multi-user environment and includes cross-sectioning and TEM prep automation options.

- Improved contrast and collection efficiency with Trinity detection and updated T2 detector
- Routine TEM sample preparation with AutoTEM™ 5 Software
- Lo-vac, cryogenic, and analytical detection configurations, including ToF SIMS



Top: Efficient cross-sectioning of interconnect structures

Bottom: TEM sample lift-out site with fiducial for robust automation

Preparation

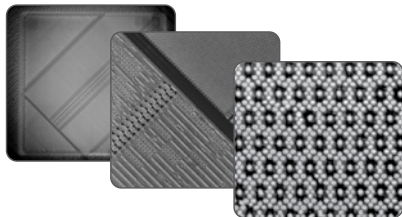
Physical Analysis (*Large-Area FIB Processing*)

Thermo Scientific™ Helios™ 5 CXe/UXe/HXe PFIB-SEM



Automated planar and non-damaging characterization of sub-5 nm logic and memory devices with stable, high resolution imaging.

- Automated multilayer end-point detection and inspection for large-area, multisite planar deprocessing
- Four-inch, high-precision Piezo UHR stage available with HXe model
- **New!** Upgrade to Helios Hydra 5+ PFIB-SEM with Damage-Free Delayering (100 eV SEM inspection and stability) and enhanced lamella preparation automation for top-down, inverted, and planar samples



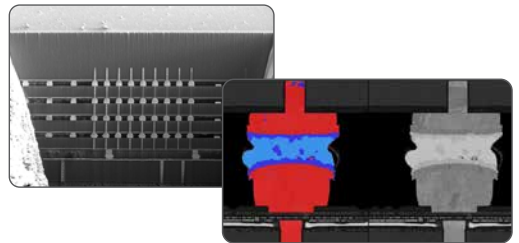
Planar delayering with 100eV SEM imaging with Helios 5+ PFIB

Thermo Scientific™ Helios™ 5 Laser PFIB-SEM



Ultra-large-volume solution with three coincident beams for millimeter-scale high-throughput laser ablation, fast polishing and precise end-pointing with PFIB, and high-resolution and multi-mode analysis with SEM and configurable analytical detectors.

- Fast characterization of subsurface features over millimeters of depth
- Accurate and repeatable cut placement with coincident beams
- Leverages all the benefits of Helios Hydra (multi-ion species) or PFIB platforms with enhanced capability for ultra-large-volume removal



Large area laser cross-section in high-bandwidth memory.

Preparation

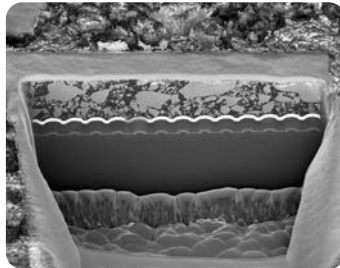
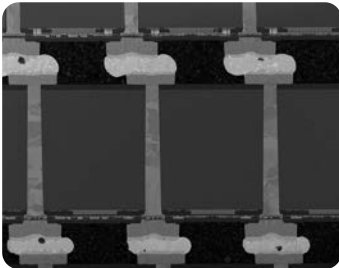
Physical Analysis (*Large-Area FIB Processing*)

Thermo Scientific™ Helios™ 5 Hydra UX/CX PFIB-SEM



Fast, selectable ion species (Xe, AR, O, N) switching for comprehensive sample preparation of complex materials and devices with increased throughput compared to traditional PFIB and optimal cut quality with optimized ion species selection.

- Switchable ion species process challenging compound and packaging materials with fast throughput and optimal cut quality
- Exceptional application flexibility for large-volume SEM cross-sectioning or 3D reconstruction, delayering, and Ga-free TEM sample preparation
- **New!** Upgrade to Helios Hydra 5+ PFIB-SEM with Extreme Milling (50% beam current increase), Damage-Free Delayering (100 eV SEM inspection and stability), and enhanced TEM prep automation



Cross-sections showing Through Silicon Vias in a packaged device and a power MOSFET, prepared with Ar⁺ ion species.

Preparation

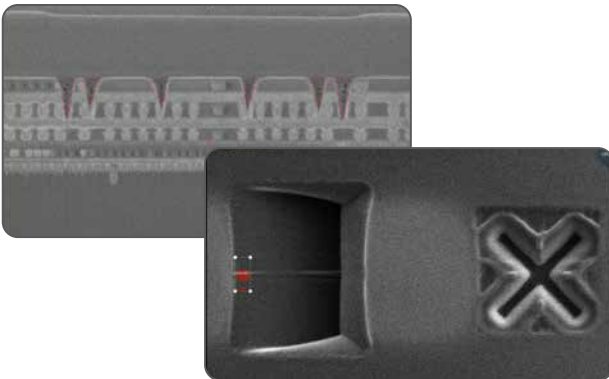
Wafer Yield Control and Metrology

Thermo Scientific™ Helios™ 5 EX/EXL Wafer DualBeam



Full wafer TEM sample preparation for metrology and defect analysis.

- Machine learning endpointing for precise cut placement
- Automated TEM sample preparation
- Low-kV precision FIB milling
- Helios 5 EX for manual wafer load, Helios 5 EXL for EFEM wafer load



Site-specific TEM sample prep and fully automated end-pointing of ROI.

Analysis

Electron microscopy is a core technology at Thermo Fisher Scientific, enabling direct inspection of nanoscale features with scanning electron microscope (SEM) and scanning transmission electron microscope ((S)TEM) solutions. We serve a broad range of semiconductor applications from automated, high-productivity metrology to high-performance tools with specialized capabilities for research and development.

The semiconductor industry is one of the most demanding, with reliable and rapid analysis supporting maximized device performance and yield. Our analytical tools are widely regarded as the process of record for leading manufacturers of logic, memory, power, analog and packaged devices. Additionally, these instruments are viewed as the reference technology in a wide variety of applications such as:

- Materials analysis
- Device debug
- Yield improvement
- Defect/failure root-cause analysis
- Metrology
- Research and development
- Ultra-high-purity gas analysis

Analysis

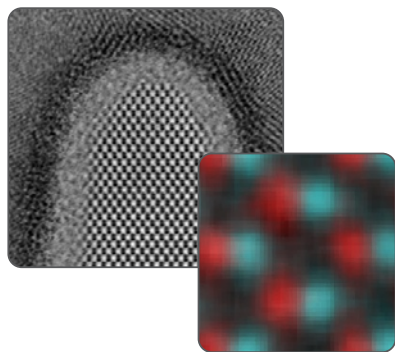
Physical Analysis (S/TEM)

Thermo Scientific™ Spectra™ Ultra (S)TEM



Atomic data for advanced semiconductor analysis and research.

- Fast, high-tension switching to tailor voltage to experiment needs
- Lowest dose STEM EDX investigation of beam-sensitive specimens
- Atomic-scale analysis of a very wide range of materials



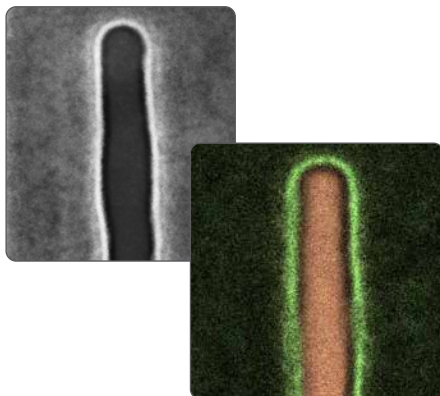
Atomic analysis of advanced logic devices.

Thermo Scientific™ Spectra™ 300/200 (S)TEM



High-resolution TEM and STEM for semiconductor applications.

- High-brightness X-CFEG
- Configuration flexibility



TEM image and EDS map..

Analysis

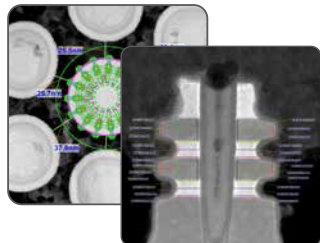
Physical Analysis (S/TEM)

Thermo Scientific™ Metrios™ 6 (S)TEM



Fully automated, high-performance, high-productivity TEM metrology and elemental analysis.

- 20% productivity improvement with automated sample loading, ROI navigation, recipe-free automation
- Fast elemental analysis with 2× greater data collection efficiency
- Enhanced data integrity with tilt accuracy and precision movement



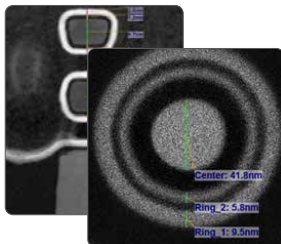
Fully-automated metrology and EDS using Thermo Scientific Ultra-X detectors.

Thermo Scientific™ Metrios™ AX (S)TEM



(S)TEM for automated and manual workflows, metrology, and elemental analysis.

- AI-driven, automated recipe generation
- Guaranteed imaging specifications
- High-volume TEM data, accurate and repeatable



Automated metrology (top) and automated EDS using Thermo Scientific Dual-X Detectors (bottom).

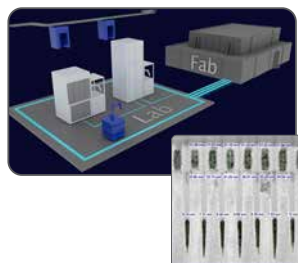
NEW!

Thermo Scientific™ Metrios™ 6S (S)TEM



Automated TEM metrology solution with operator-free TEM grid handling and seamless data connectivity.

- Part of the Vulcan Automated Lab workflow
- Fully automated TEM imaging and metrology
- Automated TEM grid handling
- Automated data connectivity



Automated TEM metrology within the Vulcan Automated Lab workflow.

Analysis

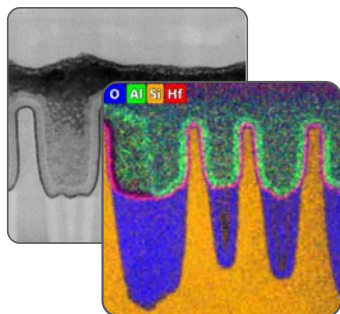
Physical Analysis (S/TEM)

Thermo Scientific™ Talos™ F200E (S)TEM



200 kV TEM and (S)TEM for repeatable, high-volume analysis of a broad range of semiconductor and microelectronic devices.

- High-quality (S)TEM imaging with low distortion
- Precise, high-speed chemical characterization
- Dedicated semiconductor-related applications



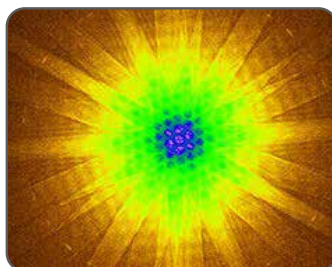
3D EDS tomogram of P-Zn-In nanotubes.

Thermo Scientific™ Talos™ F200i (S)TEM



High-productivity TEM and STEM analysis for high-quality imaging and high-throughput elemental mapping.

- Highly compact (S)TEM design for floorplan flexibility
- Designed for multi-user environment and remote operation
- Available with a wide range of high-resolution field emissions guns (FEG) including S-FEG, high-brightness X-FEG, or ultra-high-brightness cold field emission gun (X-CFEG)



Converged beam electron diffraction (CBED) pattern generated in STEM mode.

Analysis

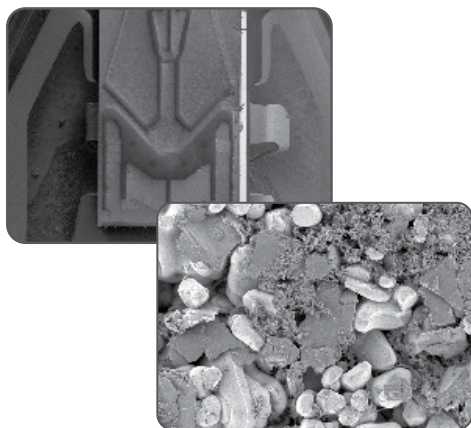
Physical Analysis (SEM)

Thermo Scientific™ Verios™ 5 XHR SEM



Precise SEM characterization of nanomaterials with sub-nanometer resolution and high material contrast.

- Automated metrology applications
- Low-dose operation and optimal contrast selection
- Sub-nanometer performance from 1 to 30kV



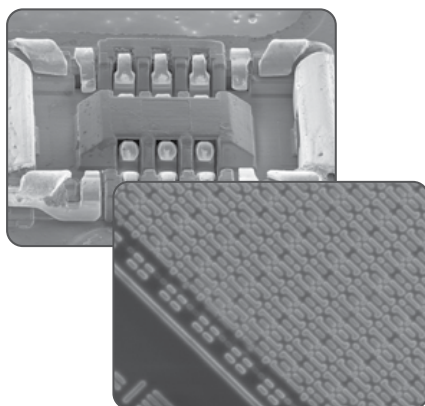
Low-voltage images of ceramic battery catalyst and hard drive reader images.

Thermo Scientific™ Apreo™ 2 SEM System



High-performance for all-round nanometer or sub-nanometer resolution.

- Extreme flexibility for handling a wide range of sample types
- Excellent performance at long working distance (10 mm)



Top: Low-voltage, low-magnification images of a component on a PCB.

Bottom: 200 V image of a deprocessed 14 nm device.

Analysis

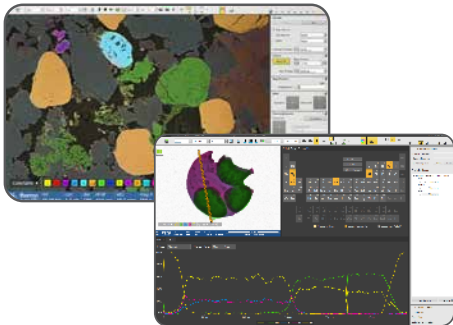
Physical Analysis (SEM)

Thermo Scientific™ Axia™ ChemiSEM



Flexible, easy-to-use and fast analytical SEM for general purpose surface analysis.

- Flexible sample loading for large, heavy samples (up to 10 kg)
- Integrated, live (ChemiSEM) EDS
- Always fully optimized with auto alignments
- Simple UI with user guidance and undo functions



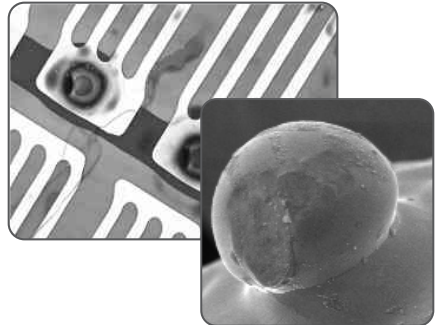
Live EDS imaging and simple user interface.

Thermo Scientific™ Phenom™ Desktop SEM



SEM product family for uncompromising performance in a desktop form factor.

- Available in a variety of configurations to suit various semiconductor applications such as particle analysis or device packaging analysis
- Resolutions of 2.5 nm (SE) and <4.0 nm (BSE) @ 15 kV
- Up to 1,000,000x magnification
- Optional EDS configurations

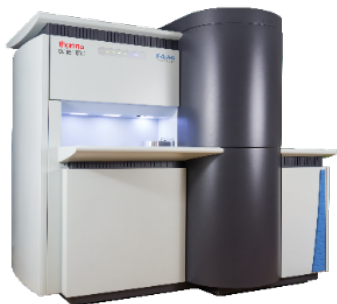


High-resolution surface SEM analysis of packaged die and electronic modules.

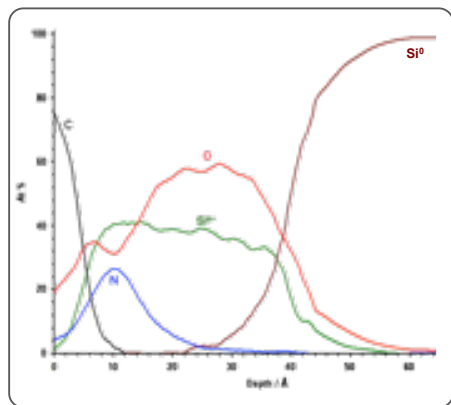
Analysis

Surface Analysis (XPS)

Thermo Scientific™ K-Alpha™ X-ray Photoelectron Spectroscopy System

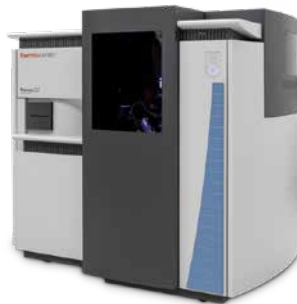


Introducing a fresh approach to surface analysis. It offers a streamlined workflow and user-friendly operation while maintaining high-quality results and advanced capabilities.

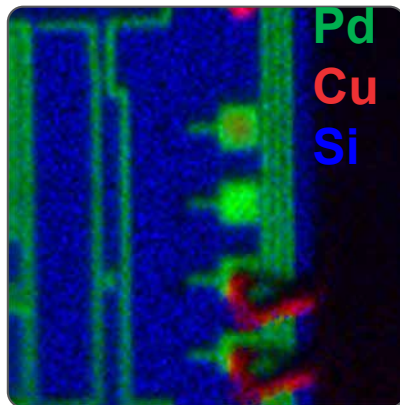


Understanding thin film structure and interface chemistry using the high-performance ion source and high-sensitivity spectrometer.

Thermo Scientific Nexsa™ G2 Surface Analysis System



Fully automated, high-throughput surface analysis. It integrates XPS with ion scattering spectroscopy (ISS), UV photoelectron spectroscopy (UPS), reflected electron energy loss spectroscopy (REELS), and Raman spectroscopy for true correlative analysis. With options for sample heating and biasing, it expands the range of possible experiments.



The small spot X-ray source enables identification of pad surface contamination, resist composition, and pad structure through depth profiling.

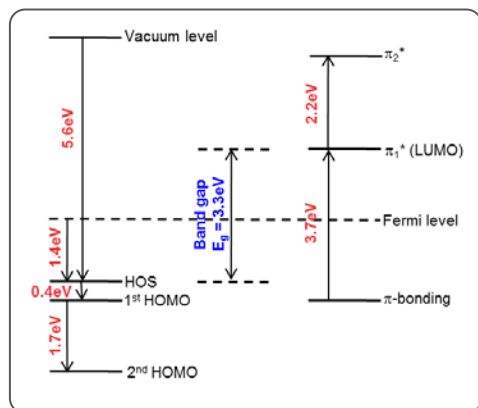
Analysis

Surface Analysis (XPS)

Thermo Scientific™ ESCALAB™ QXi X-Ray Photoelectron Spectroscopy Microprobe



Highly flexible and configurable multi-technique instrument, offering exceptional sensitivity and rapid production of high-quality spectra. The system features a unique dual detector system that enables superb XPS imaging with excellent spatial resolution. XPS, ISS, and REELS are included as standard, and the instrument can be configured with optional HAXPES, AES, UPS, Raman, and IPES methods. Thermo Scientific Avantage Software seamlessly integrates system control, data acquisition, processing, and reporting.



XPS, REELS and UVPS can be used together to understand work function and band gap properties.

Analysis

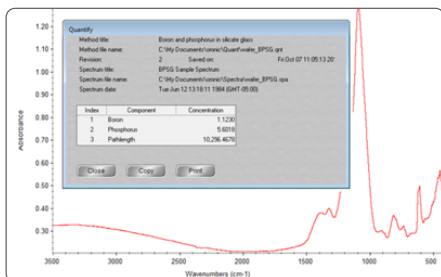
Chemical Analysis (*RAMAN and FTIR*)

Thermo Scientific™ Nicolet™ iS50 FTIR Spectrometer



Reliable QC and characterization of:

- Epitaxial film thickness
- Interstitial O and substitutional C levels in wafers
- Dielectric film characterization

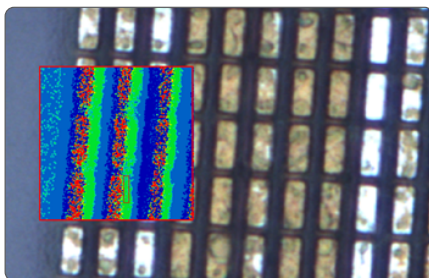


FTIR spectral analysis of BPSG film.

Thermo Scientific™ DXR™ 3xi Raman Imaging Microscope



- Chemical composition of compound semiconductors
- Crystal structure type, orientation and polymorphs
- Structure, stress, defects and treatments of semiconductor materials and film
- Dopant concentrations



Raman imaging analysis of strained regions in silicon.

Analysis

Chemical Analysis (*RAMAN and FTIR*)

Thermo Scientific™ Nicolet™ RaptIR™+ FTIR Microscope



- Visual spatial resolution of 1 micron and infrared spatial resolution of 5 microns
- Analyze uniformity of photoresist
- Analyze residues after processing
- Analyze particulate contaminants from fab

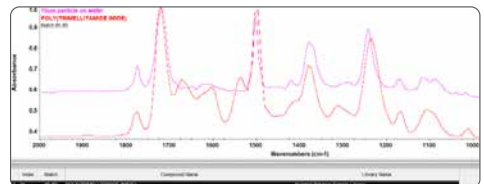
Thermo Scientific™ Nicolet™ iN10 Infrared Microscope



- Room temperature detector for samples down to 50 microns and MCT array for high spatial resolution and fast imaging
- Analyze Epi thickness uniformity in wafers
- Failure analysis with contact ATR method or non-contact reflection and transmission analysis



Particle contamination on surface of a wafer component causing failure



Analyze the spectra and identify the particle and source of contamination

Analysis

Chemical Analysis (*IC, ICP-MS, GC-MS*)

Thermo Scientific™ Dionex™ ICS 6000+ IC



- QA/QC of UPW and chemicals
- Anions cations contamination analysis

Thermo Scientific™ iCAP™ MX Series ICP-MS



- QA/QC, UPW and semiconductor-grade chemicals
- Ultratrace elemental impurities

Thermo Scientific™ ISQ™ 7610 Single Quadrupole GC-MS



- QA/QC of CR air and gases
- Organic contamination analysis

Thermo Scientific™ Element™ Series High-resolution ICP-MS



- QA/QC and research
- Interference-free ultratrace impurities

Analysis

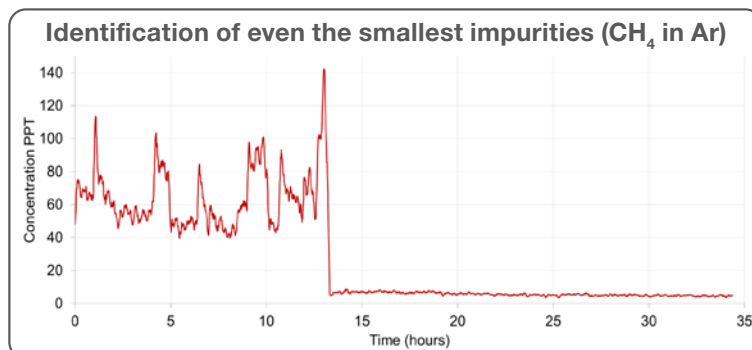
Gas Purity Analysis (*APIX*)

Thermo Scientific™

APIX™ dQ API-MS High Purity Gas Analyzer



- Parts-per-trillion detection of impurities in utility gases
- Automated and online 24/7 multi-stream analyzer
- Protect valuable wafers from defects due to contaminants



Repair to gas handling equipment demonstrates the value of API-MS detection capability.

Analysis

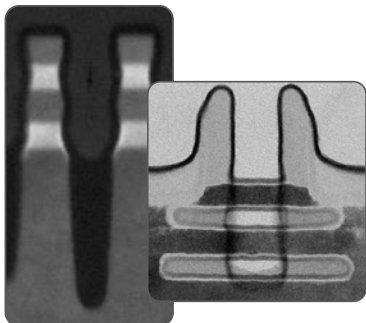
Wafer Yield Control and Metrology

Thermo Scientific™ Helios™ 5 EX/EXL Wafer DualBeam



FIB-SEM TEM sample preparation for semiconductors, enabling full-wafer analysis.

- Machine learning endpointing for precise cut placement
- Automated TEM sample preparation
- Low-kV precision FIB milling
- Helios 5 EX for manual wafer load, Helios 5 EXL for EFEM wafer load



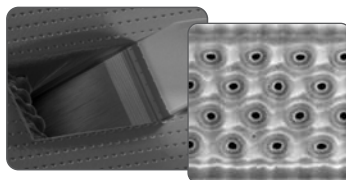
TEM of a Si/SiGe gate-all-around (GAA) fin (left), TEM along the nanowire of a GAA structure (right).

Thermo Scientific™ Helios™ 5 PX/PXL Wafer DualBeam



PFIB SEM for in-line metrology and process monitoring of advanced 3D NAND, DRAM and wafer-level packaging applications.

- Fast, precise, large-area, wafer-level deprocessing, diagonal milling and cross sectioning of high-aspect-ratio structures
- Advanced automation capabilities accelerates process development
- High-resolution, high-contrast imaging for exact dimensional measurements of high-aspect-ratio 3D through-stack structures
- Helios 5 PX for manual wafer load, Helios 5 PXL for EFEM wafer load



Wide-area diagonal mill of 3D NAND device (left) and SEM inspection of diagonal unfilled structures on same (right).

Analysis

Wafer Yield Control and Metrology

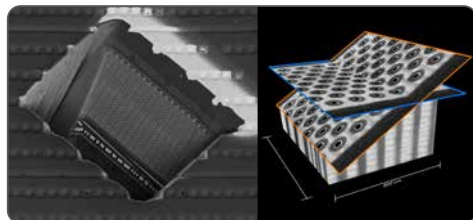
NEW!

Thermo Scientific™ Helios™ MX1 PFIB-SEM



Fully automated, multi-ion species wafer PFIB system that delivers precise 3D SEM metrology and buried defect analysis of 3D NAND, DRAM, logic, and wafer-level packaging applications.

- Rapid, accurate, wafer-scale delayering, diagonal milling, and cross-sectioning of high-aspect-ratio structures over large areas
- Multi-ion species FIB, including argon for fast, large-volume milling
- Integrated workflows for 3D reconstruction and metrology
- Enhanced throughput with Metrology Mode image acquisition



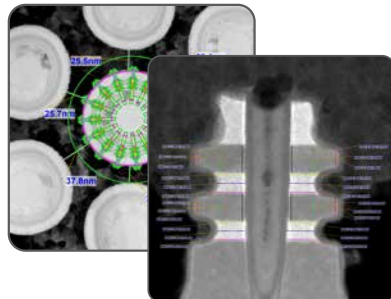
Automated 3D NAND metrology using diagonal mill

Thermo Scientific™ Metrios™ 6 (S)TEM



Fully automated, high-performance, high-productivity TEM metrology and elemental analysis.

- 20% productivity improvement with automated sample loading, ROI navigation, recipe-free automation
- Fast elemental analysis with 2× greater data collection efficiency
- Enhanced data integrity with tilt accuracy and precision movement



Fully-automated metrology and EDS using Thermo Scientific Ultra-X detectors.

Analysis

ESD Compliance Testing

**Thermo Scientific™
MK™ Series**



Automated ESD testing to industry standards for human body model, machine model and latch-up.

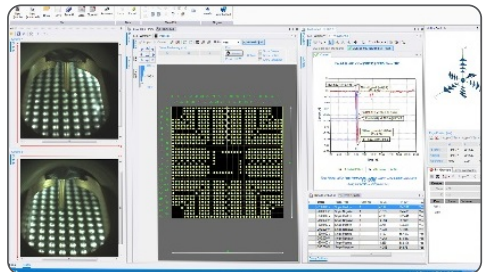
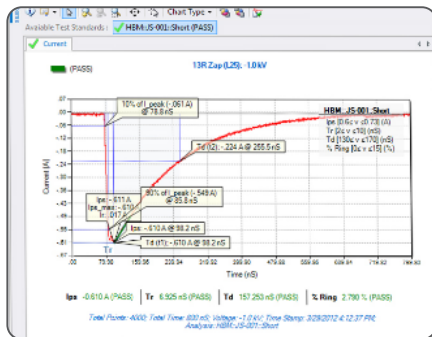
- Rapid-relay-based operations
- Latch-Up stimulus and device biasing
- Easy-to-use operations

**Thermo Scientific™
Orion3™ System**



Charged device model discharge testing for simulating electrostatic discharge events.

- Supports all popular test standards
- Dual high-resolution cameras
- Event detection and automatic waveform capture



Analysis

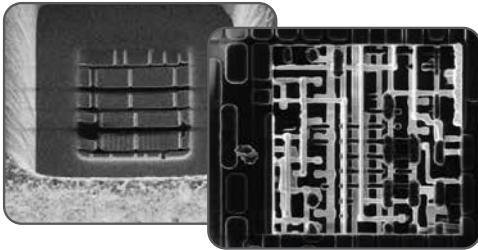
Circuit Edit

Thermo Scientific™ Centrios™ HX System



Precision circuit editing for advance technology node debug, repair and prototyping.

- Cutting-edge performance for sub-7 nm advanced semiconductor
- 25% higher resolution at 4x lower beam currents for unparalleled milling precision and control with low landing energy
- Innovative dual-nozzle gas-delivery system with a broad portfolio of chemistries, enabling high-precision etching and fast, efficient editing



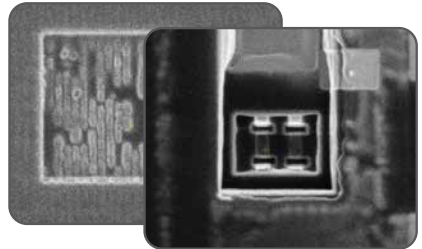
Featuring the new Thermo Scientific Celta Focused Ion Beam (FIB) Column and providing incredible control for performing edits accurately and with minimum circuit damage.

Thermo Scientific™ Centrios™ CX System



Advanced circuit editing for semiconductor debug, repair and prototyping.

- Imaging and milling resolution to meet 14 nm node applications
- Advanced front and backside edits with exceptional editing control and precision
- Planarity/uniformity, delayering/etch stopping and high-acuity drilling



High-resolution Tomahawk WDR FIB opens low-level metal layer with precision and control, while Dual Multichem design enables repeatable, precise and uniform process control.

Analysis

Circuit Edit

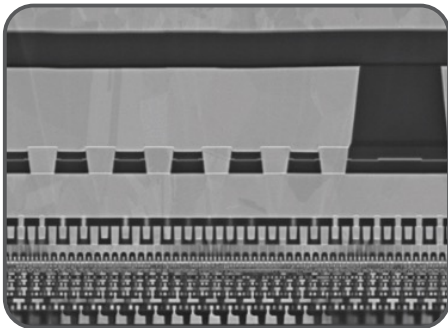
NEW!

**Thermo Scientific™
Centrios™ JX System**



Our most advanced circuit edit solution, designed to address challenges introduced by backside power rail technology.

- ICE+ detector enhances collection efficiency, resolving SNR loss associated with backside circuit edit applications
- UHR+ stage enables superior macro-navigation, with up to 2x precision improvement over UHR stage
- Beam deflection navigation delivers local accuracy below 2.5nm
- Versatile applications including frontside and backside edits for device prototyping and repair and sample preparation for eBeam probing



In backside power distribution (BPD) architecture, limited access and signal from high aspect ratio front and backside edits are improved with ICE+ detection and high precision beam control.

Automated Workflows

Near-Line Metrology

Calibrated and distortion-free imaging is essential for precise and accurate metrology, enabling fabrication engineers to make mission-critical process decisions. The ability to produce large amounts of quality automated data is fundamental to understanding how 2D and 3D structures evolve from the front to back end of the line. This data is equally important as reference metrology for optical critical dimension (OCD) models. Automated imaging and metrology enable fast, consistent and precise data at a fraction of the operator overhead. Our enhanced throughput workflows are engineered for industry leading productivity and fast time-to-data and are the process of record across the semiconductor world.

Root-Cause Analysis

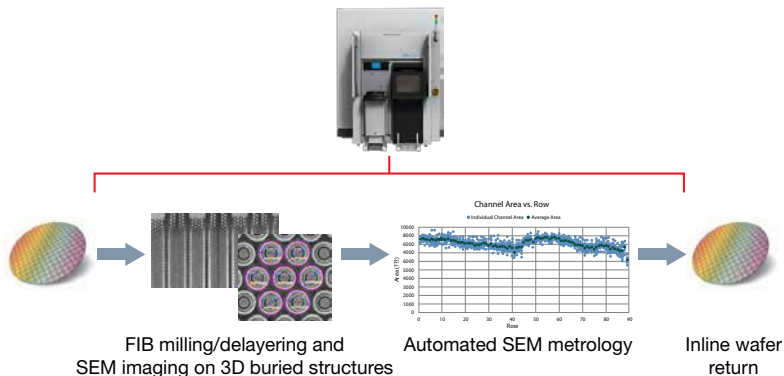
The management and elimination of defects during fabrication are becoming more and more challenging as they become smaller and buried in relatively large 3D structures. Thermo Fisher Scientific provides workflow solutions to locate, isolate and study defects on a broad range of device types and scale lengths. We have developed high-yield, high-productivity solutions that can work from the millimeter to atomic scale and provide comprehensive chemical, structural and physical information, all with fast time-to-data.

Automated Workflows

SEM and TEM Metrology

3D SEM Metrology

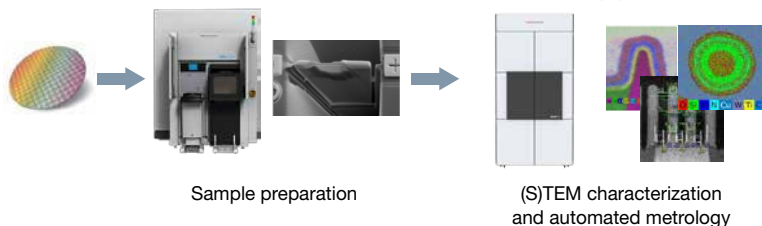
Helios PFIB-SEM Wafer DualBeam



S/TEM Metrology

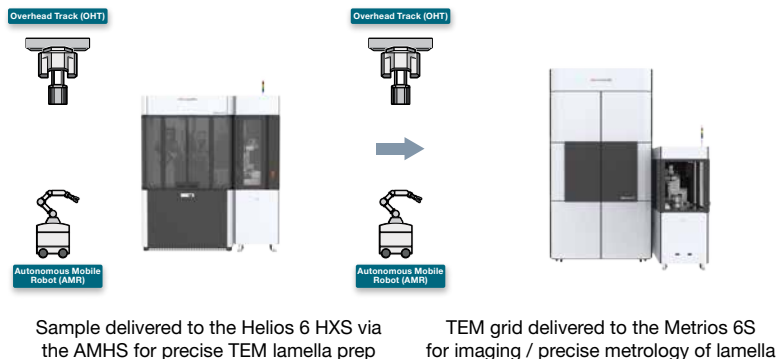
Helios 5 EXL FIB-SEM Wafer DualBeam

Metrios 6 (S)TEM



NEW!

Vulcan Automated Lab



Automated Workflows

Root-Cause Analysis



Helios Hydra 5+ FIB-SEM

Precise, site-specific, automated delayering with damage-free 100 eV inspection and high-throughput, large-area, cross-section imaging.

nProber NanoProbing

Simple, reliable fault isolation and transistor characterization at the nanometer scale.

Helios FIB-SEM

High-productivity tools for cross-section imaging and analysis for TEM and atom probe sample preparation; available in small sample.

Metrios/Spectra/Talos (S)TEM

Fast time to data on TEM platforms that can be configured for a variety of defect analysis applications; for example, S/TEM imaging, EDS, EELS, strain.

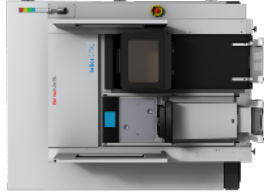
Automated Workflows

Root-Cause Analysis

NEW!

Fully-automated workflow for Voltage Contrast (VC) defect redetection and root cause analysis

Helios 5 EXL

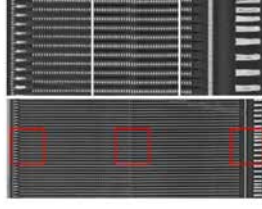


VC Redetection



Automatic

Final Cross-section
+ SEM Image



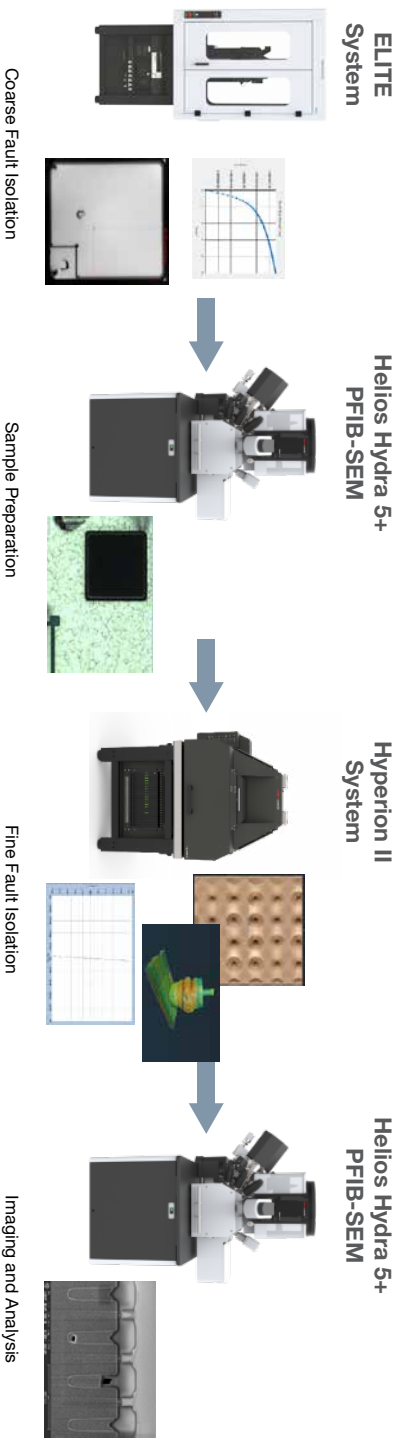
Incoming wafer from
defect inspection system

Helios 5 EXL uses VC redetection software to re-locate defects
and precisely create cross sections revealing the buried defect

Next, a final high-resolution SEM image
is taken, allowing the root cause of
the defect to be determined

EFA to PFA Workflows

Example: FA on Power MOSFET Devices



Precision sample preparation + high-accuracy fault localization = high-performance, high-success-rate PFA

Software Solutions

Software with robust capabilities plays a crucial role in semiconductor analysis. Software automates various tasks, such as navigation to specific regions of interest, image processing and data analysis, and reduces the time and effort required by operators. Software provides tools for data visualization, enabling analysts to explore critical features and analyze and understand increasingly complex 3D structures. Software improves analysis accuracy by minimizing human error and enhancing the detection of subtle defects or failure mechanisms. Thermo Fisher Scientific provides innovative software solutions to streamline processes, enhance accuracy, and enable deeper insights into semiconductor devices leading to better-informed decision-making and increased productivity.



CAD Navigation

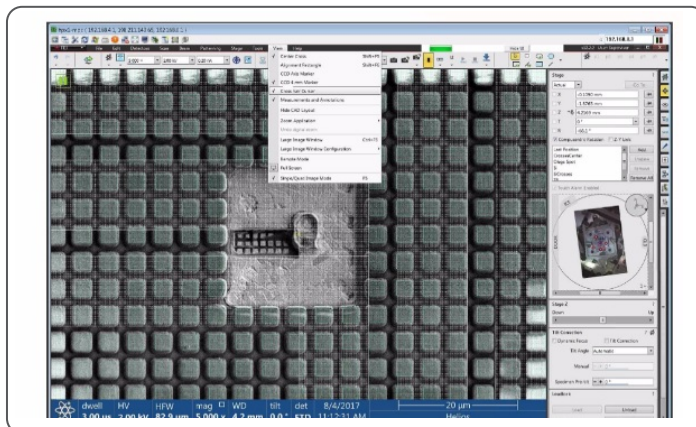
CAD-to-Stage Navigation Capabilities

Thermo Scientific™ NEXS™ Software goes far beyond a CAD viewer and stage driver by facilitating fault isolation, failure analysis, sample preparation, and circuit edit. NEXS Software enables diverse workflows by providing CAD connectivity with Thermo Scientific analytical equipment.

- Easy-to-use CAD viewing automatically drives the system stage to the precise ROI indicated by the CAD model.
- Reads mask data from GDS2 and OASIS format files.



Provides user-friendly navigation to CAD coordinates on all Thermo Scientific product platforms (circuit editing, DualBeams, Meridian Fault Isolation Systems, etc.).



Sample navigation using CAD overlay on an image from a Thermo Scientific Helios DualBeam.

Avizo Software for Semiconductor

From fundamental semiconductor materials research to automated electronic components quality analysis

Thermo Scientific™ Avizo™ Software is a solution for data interpretation that enables complex metrology challenges on 2D, 3D and multidimensional data.

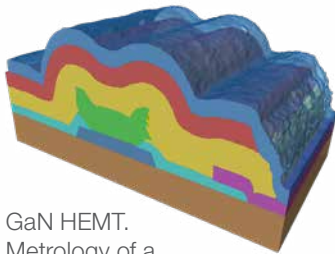
Avizo Software grants you access to target sample parameters with numerical quantification relying on AI segmentation and image enhancement capabilities, robust methods for data reconstruction and multidimensional visualization.

Examples of how some users benefit from Avizo Software:

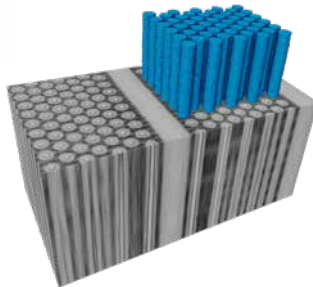
- Simplification of quality control and failure analysis tasks with automated image processing workflows and reporting
- Visualization and exploration of multi-channel imaging and spectroscopy datasets
- Measurement of relevant materials parameters (porosity, tortuosity) or detect contaminants embedded in the material

Avizo Software can process most imaging formats independently from the source. Computer tomography or electron microscopy data are just some examples of the supported acquisition technologies.

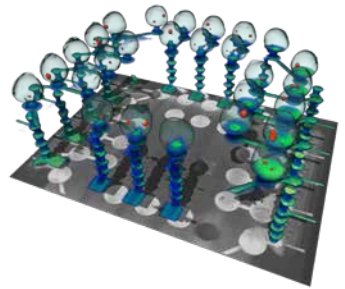
Avizo Software and customization services help ensure we cover the needs of each user with off-the-shelf solutions or workflows that are completely customized for their tasks.



GaN HEMT.
Metrology of a
transistor. Imaged with
a Thermo Scientific
Helios 5 HX DualBeam



3D NAND. Memory channels
segmented for CD analysis.
Imaged with a Thermo Scientific
Helios 5 PXL PFIB DualBeam



Ball grid array showing
solder porosity. Imaged
with micro-CT. *Image
courtesy of nanoX
Technology Pte. LLtd.*

Semiconductor Service Solutions

Integrated Service Solutions to Drive Your Business

Our technology-driven, expert care will help you:

- Increase system productivity
- Avoid unplanned downtime
- Upskill and empower users
- Improve fleet performance

Our service offerings keep your system in peak condition with the latest software updates, expert telephone support, RAPID remote diagnostics, and regular preventive maintenance.

Our Accelerate and Advance portfolios transform service with a data-driven approach so you can proactively plan service activities and align your fleet for optimal operations. The Connected Care Portal puts data at your fingertips so you can monitor system utilization and trends.

Service Offereing	Warranty	Classic Contracts	Accelerate & Advance Contracts
Phone support	•	•	•
Preventive Maintenance	•	•	•
Software updates	•	•	•
Parts	•	•	•
RAPID Connection	•	•	•
On-site response time	48 hours	4–72 hours	4–48 hours
Connected Care Portal			•
System Remote Monitoring			•
Uptime guarantee			•
After hours on-site service			•
Regular account reviews			•
Fleet compare			•
Consumable monitoring			•

Specific service offerings vary across contracts and geographies.

Perform Services

Productivity and System Performance

Our Perform offerings (part of our Classic Services) are comprehensive in critical system support, boosting performance, minimizing downtime, and helping you achieve consistent business outcomes.

- Quickly connect to system experts via the web, email or phone for troubleshooting and system inquiries
- Keep systems in peak condition with the latest software updates, including license and installation
- Get Thermo Scientific quality parts expedited on-site from strategic global inventories
- Ensure a rapid response with priority access to Thermo Fisher engineers
- Experience optimum system function with regular preventive maintenance

In addition to our technical experts, software updates and access to RAPID remote service and telephone support, our Perform service options offer a wide range of features that provide the support you need to deliver exceptional results.

	Perform Pro	Perform Assure	Perform Complete	Perform Enhanced	Perform Edge	Perform Ultimate
On-site response	72-hour	48-hour	48-hour	24-hour	4-hour	4-hour + weekend on-call coverage
Parts	15% discount	15% discount CM parts	•	•	•	•
Consumables	PM consumables only		Optional	Optional	Optional	Optional
Extended FSE hours				•	•	•
Account reviews				Semi-annual	Monthly	Weekly
Local parts stocking					Optional	Optional
Targeted system uptime					•	
Guaranteed system uptime						•

Accelerate & Advance Services

Transforming service with a data-driven approach

As a part of the competitive semiconductor and electronics industry, you need to be able to rely on your equipment to meet the ongoing need for uptime and results to support your company's aggressive goals.

With Accelerate and Advance Services, we offer an end-to-end solution that integrates convenient access to data, continuous system remote monitoring, the support of a Customer Success Manager, and the comprehensive on-site maintenance you are accustomed to.

Customers with Accelerate and Advance experience up to 10% fewer unplanned downtime hours.

Remote System Monitoring

- Critical parameters are monitored
- Proactive service engagement if anomalies are detected

Connected Care Portal

- System utilization and trends
- Average uptime and trends
- Fleet Compare
- Digital Reporting

Optimize System Performance

- Guaranteed uptime
- Reduce unplanned downtime
- After-hours service support

Maximize Productivity

- Customer Success Manager to drive customer objectives via entitlements
- Applications support that focuses on system reliability and consistency



Product Table

Thermo Scientific Systems

	Page #	Localization				Preparation					Analysis									
		Thermal Fault Isolation	Optical Fault Isolation	E-Beam Probing	Nanoprobing	Laser X-Sectioning	Delayering	FIB X-Sectioning	TEM Sample Preparation	Atom Probe Preparation	Circuit Edit	S/TEM	SEM	Metrology	Elemental Analysis	Defect Analysis	Chemical Analysis	Gas Purity Analysis	Surface Analysis	ESD Testing
Thermo Scientific™																				
Elite™ System	6	■																		
Meridian™ System	6		■																	
Meridian™ EX System	6			■																
Meridian™ S System	7		■																	
nProber™ System	7				■															
Hyperion™ II System	7				■															
Helios™ 6 HD FIB-SEM	9						■	■				■			■	■				
Helios™ 6 HX/HXS FIB-SEM	9						■	■				■			■	■				
Helios™ 5 FX DualBeam	10						■	■	■			■			■	■				
Helios™ 5 UC/CX DualBeam	10						■	■				■			■	■				
Scios™ 3 DualBeam	10						■	■				■			■	■				
Helios™ 5 PFIB CXe/UXe/HXe DualBeam	11						■	■	■			■			■	■				
Helios™ 5 Laser PFIB-SEM	11					■	■	■	■			■			■	■				
Helios™ Hydra 5 UX/CX PFIB-SEM	12						■	■	■			■			■	■				
Helios™ Hydra 5 Laser PFIB-SEM	11					■	■	■	■			■			■	■				
Helios™ 5 EX/EXL Wafer DualBeam	13, 26							■	■			■			■	■				
Helios™ 5 PX/PXL Wafer DualBeam	26						■	■	■			■	■		■	■				
Helios™ MX1 PFIB-SEM	27						■	■							■	■				
Talos™ F200E / F200i (S)TEM	17											■		■	■	■				
Spectra™ Ultra (S)TEM	15											■			■	■				
Spectra™ 300/200 (S)TEM	15											■			■	■				
Metrios™ 6/6S (S)TEM	16, 27											■		■	■	■				
Metrios™ AX (S)TEM	16											■		■	■	■				
Verios™ 5 XHR	18												■	■	■	■			■	
Phenom™ Desktop SEM	19												■		■	■			■	
Axia™ ChemiSEM	19												■		■	■			■	
Apreo™ 2 System	18												■		■	■			■	
MK Series	28																			■
Orion™ 3 System	28																			■
Centrios™ HX System	29									■										
Centrios™ CX System	29									■										
Centrios™ JX System	30									■										
K-Alpha™ XPS System	20														■				■	
Nexsa™ G2 XPS System	20														■				■	
ESCALAB™ QXi+ System	21														■				■	
Nicolet™ iS50 FTIR Spectrometer	22																■			
Nicolet™ RaptIR™+ FTIR Microscope	23																■			
Nicolet™ iN10 Infrared Microscope	23																■			
DXR™ 3xi Raman Imaging Microscope	22																■			
Dionex™ ICS 6000+ IC	24																■			
iCAP™ MX Series ICP-MS	24																■			
iSQ™ 7610 Single Quadrupole GC-MS	24																■			
Element™ Series High-resolution ICP-MS	22																■			
APIX™ dQ API-MS High Purity Gas Analyzer	25																	■		

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

REVEAL what matters **ENABLE** what's critical **INNOVATE** what's next

With over 40 years of experience,
and a fast, responsive global support
network, we are your trusted partner
for semiconductor analysis.

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